

DeepThinkTrader Setup Guide

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CRITICAL WARNING: Start with paper trading ONLY. Never use real money until you have backtested for 3+ months. Trading bots can lose 100% of capital. This project is educational and experimental.

1. Required API Keys

You need three free API accounts. All keys go in the `.env` file (never commit this file).

1.1 Alpaca Markets (Paper Trading)

- 1. Go to **alpaca.markets** and create a free account
- 2. Navigate to **Paper Trading** dashboard
- 3. Click **API Keys** → **Generate New Key**
- 4. Copy both the **API Key ID** and **Secret Key** (secret shown only once!)

Variable	Description	Example
ALPACA_API_KEY	Your Alpaca API Key ID	PK... (20 chars)
ALPACA_SECRET_KEY	Your Alpaca Secret Key	abc123... (40 chars)
ALPACA_BASE_URL	Paper trading endpoint	https://paper-api.alpaca.markets

1.2 NewsAPI

- 1. Go to **newsapi.org** and register for free
- 2. Copy your API key from the dashboard
- 3. Free tier: 500 requests/day (sufficient for hourly research on 5 tickers)

Variable	Description	Example
NEWSAPI_KEY	Your NewsAPI key	abc123def456... (32 chars)

1.3 Reddit API (PRAW)

- 1. Log into Reddit, go to reddit.com/prefs/apps
- 2. Click "create another app..." at the bottom
- 3. Select "script" type
- 4. Name: DeepThinkTrader
- 5. Redirect URI: http://localhost:8080
- 6. Click Create app
- 7. Note the client ID (under the app name) and secret

Variable	Description	Example
REDDIT_CLIENT_ID	App client ID (under app name)	aBcDeFgHiJkLmN (14 chars)
REDDIT_CLIENT_SECRET	App secret	xYzAbCdEfGhIjKlMnOpQrStUvWx (27 chars)
REDDIT_USER_AGENT	Identifies your app to Reddit	DeepThinkTrader/1.0

2. Trading Parameters

All configurable via `.env` file. Defaults are conservative for paper trading.

Parameter	Variable	Default	Description	Recommended Range
Account Size	ACCOUNT_SIZE	\$50,000	Starting paper balance	\$10K - \$100K
Watchlist	WATCHLIST	NVDA,TSLA,AAPL,AMD,SPY	Comma-separated tickers	3-10 tickers
Max Risk/Trade	MAX_RISK_PER_TRADE	0.02 (2%)	Max % of account risked per position	0.01 - 0.03
Max Daily Loss	MAX_DAILY_LOSS	0.05 (5%)	Hard stop — no more trades today	0.03 - 0.06
Min Conviction	MIN_CONVICTON	8	Score threshold to execute (1-10)	7 - 9
Research Interval	RESEARCH_INTERVAL_MINUTES	60	Minutes between research cycles	30 - 120
Min R:R Ratio	MIN_REWARD_RISK_RATIO	2.0	Take-profit must be 2x stop-loss	1.5 - 3.0

3. Risk Management Rules (Hardcoded)

These safety checks run BEFORE every trade and cannot be overridden via configuration.

Rule	Description	Action if Triggered
Conviction Gate	Analysis conviction must be $\geq$ MIN_CONVICTION	HOLD — no trade
Position Risk Cap	Max loss per position $\leq$ MAX_RISK_PER_TRADE	BLOCKED
R:R Minimum	Take-profit $\geq$ MIN_REWARD_RISK_RATIO $\times$ stop-loss	BLOCKED
Daily Loss Limit	Realized P&L today must be within MAX_DAILY_LOSS	BLOCKED for rest of day
Max Open Positions	No more than 5 simultaneous positions	BLOCKED
No Duplicates	Cannot open second position in same ticker	BLOCKED
Market Hours	Must be within US market hours (9:30-4:00 ET)	BLOCKED
Revenge Trading	3+ consecutive losses triggers cooldown	BLOCKED + warning

4. Quick Setup Checklist

#	Step	Done?
1	Create Alpaca paper account and get API keys	☐
2	Get NewsAPI key (free tier)	☐
3	Create Reddit app and get PRAW credentials	☐
4	Copy <code>.env.template</code> to <code>.env</code> and fill in keys	☐
5	Create Python venv and install requirements	☐
6	Run <code>python main.py once</code> to test single cycle	☐
7	Run <code>streamlit run dashboard.py</code> to verify dashboard	☐
8	Start scheduled loop: <code>python main.py</code>	☐

5. .env File Template

```
# Alpaca API (Paper Trading)
ALPACA_API_KEY=your_alpaca_api_key
ALPACA_SECRET_KEY=your_alpaca_secret_key
ALPACA_BASE_URL=https://paper-api.alpaca.markets

# NewsAPI
NEWSAPI_KEY=your_newsapi_key

# Reddit (PRAW)
REDDIT_CLIENT_ID=your_reddit_client_id
REDDIT_CLIENT_SECRET=your_reddit_client_secret
REDDIT_USER_AGENT=DeepThinkTrader/1.0

# Trading Parameters
ACCOUNT_SIZE=50000
WATCHLIST=NVDA,TSLA,AAPL,AMD,SPY
MAX_RISK_PER_TRADE=0.02
MAX_DAILY_LOSS=0.05
MIN_CONVICTION=8
RESEARCH_INTERVAL_MINUTES=60
MIN_REWARD_RISK_RATIO=2.0
```

DeepThinkTrader v1.0 — Paper Trading Mode Only

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